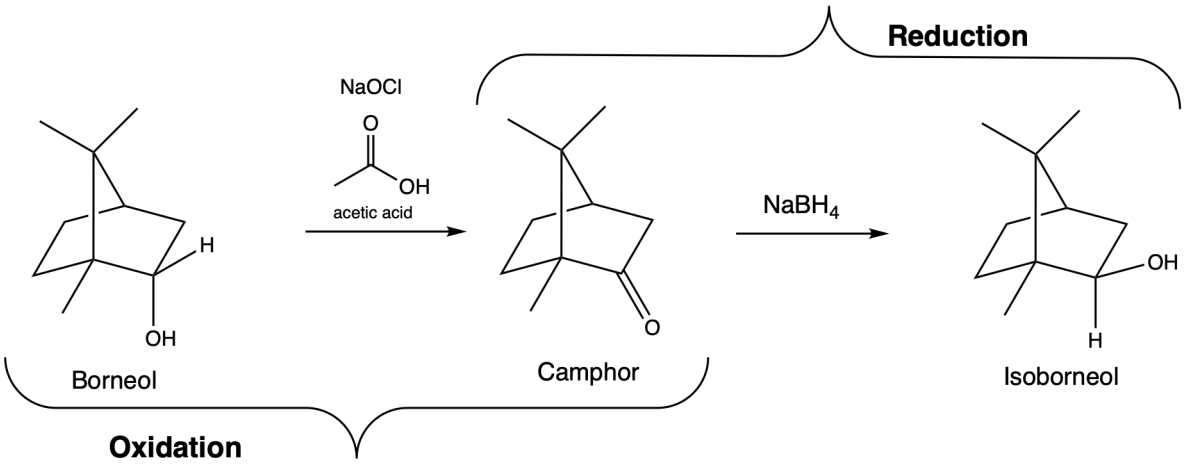
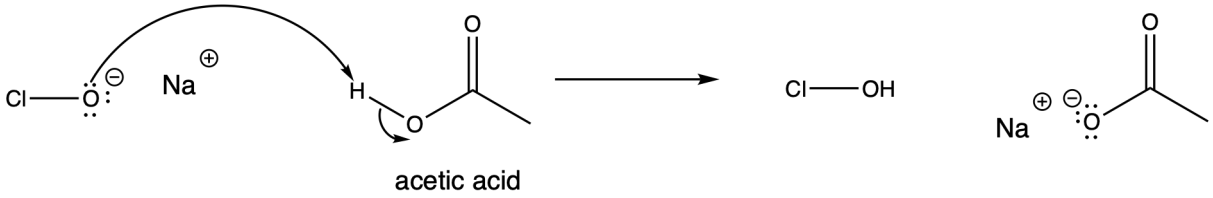
An Oxidation-Reduction Multi-Step Reaction Sequence: Borneol, Camphor, Isoborneol*



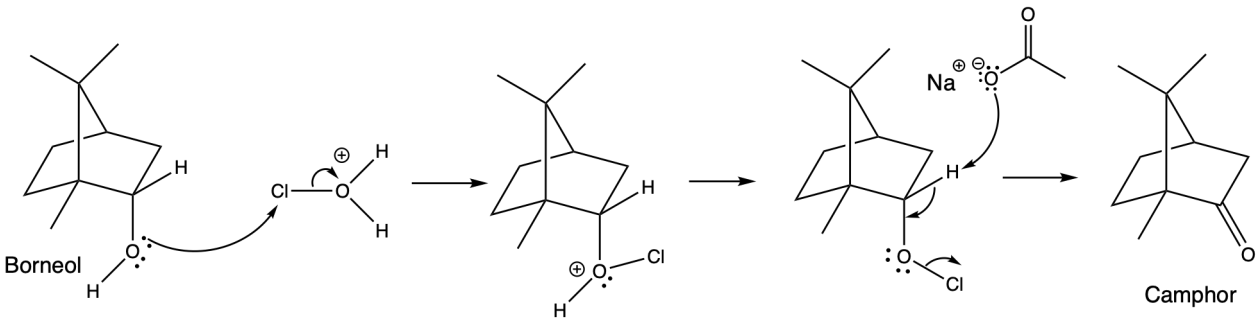
In this multi-week experiment, the student will oxidize borneol to camphor and then reduce camphor to isoborneol. The experiment entails carrying out the reaction procedures and subsequently purifying and characterizing the products.

OXIDATION OF BORNEOL WITH HYPOCHLORITE

Sodium hypochlorite, commonly known as bleach, can be utilized for oxidizing secondary alcohols into ketones. This oxidation reaction proceeds more efficiently in an acidic medium, suggesting that the primary oxidizing species is hypochlorous acid (HOCl). The active oxidizing agent is formed through the reaction between sodium hypochlorite and acetic acid. Oxidation with bleach is considered more green than many of the alternative oxidizing agents [https://reagents.acsgcipr.org/reagent-guides/oxidation-to-aldehyde-and-ketones/list-of-reagents/naocl-bleach-oxidation]



Altough the mechanism is not fully understood, evidence suggests the production of an alkylhypochlorite intermediate. A helpful way to envision the reaction might be:



* Adapted from

https://www.as.wvu.edu/~jpenn/Chem%20339/Experiments/Exp%2012%20Oxidation-Reduction%20Scheme%20for%20Borneol%20to%20Camphor%20to%20Isoborneol.pdf
and

Pavia, D.L.; Lampman, G.M.; Kriz, G.S.; Engel, R.G. Introduction to Organic Laboratory Techniques, 3rd Edition, **1990**, Saunders College Publishing, New York, NY.

Week 1 Oxidation of Borneol to Camphor

Prepare reaction mixture:

1. Add 0.360 g of racemic borneol to a 10-mL round-bottom flask.
2. Add 1.0 mL of acetone.
3. Add 0.30 mL of glacial acetic acid.
4. Attach a magnetic stir bar.
5. Attach an air condenser.
6. Place the flask in a 45°C water bath, maintaining temperature between 40-50°C. *IMPORTANT*
7. Stir until borneol dissolves; add 0.50 mL acetone if necessary.
8. While stirring, add 6.0 mL of sodium hypochlorite solution dropwise over 35 minutes through the air condenser.
9. Stir the mixture for an additional 15 minutes after addition is complete.

Workup:

1. Cool mixture to room temperature.
2. Remove air condenser.
3. Transfer mixture to a separatory funnel.
4. Add 2.0 mL t-butyl methyl ether to extract camphor.
5. Cap separatory funnel and shake, venting occasionally.
6. Transfer the lower aqueous layer into small beaker or Erlenmeyer flask.
7. Transfer the upper t-butyl methyl ether layer into another small beaker or Erlenmeyer flask.
8. Extract remaining product from aqueous layer with another 2.0 mL portion of t-butyl methyl ether, Transfer lower aqueous layer to a small beaker or Erlenmeyer flask. Collect the upper layer in the previously collected t-butyl methyl ether solution. Now all the t-butyl methyl ether should be in the beaker or flask.
9. Set aside the aqueous layer into the back of the hood.
10. Put the combined t-butyl methyl ether layers in the separatory funnel and wash with 2.0 mL saturated sodium bicarbonate solution.
11. Remove the aqueous layer; leave the organic layer in the separatory funnel.
12. To the remaining t-butyl methyl ether layer in the separatory funnel, wash with 2.0 mL 5% sodium bisulfite, remove aqueous layer.
13. Wash the remaining t-butyl methyl ether layer with 2.0 mL water, transfer, remove aqueous layer.
14. Transfer the t-butyl methyl ether layer to an Erlenmeyer flask.
15. Add granular anhydrous sodium sulfate until t-butyl methyl ether solution is dry.
16. Filter into a weighed labeled vial.
17. Evaporate until solid remains.

Week 2: Analysis of Camphor

1. Weigh the flask to determine product weight and calculate percentage yield.
2. Determine melting point; pure racemic camphor melts at 174°C.
3. Record the IR and NMR spectra of the sample.